



WTVB05 Product Data Sheet & User Manual



Product Specifications:

Model: WT-VB05

Description: Vibration sensor

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1. Product Overview

1. The module has its own voltage stabilization circuit, the operating voltage is 5V~36V, and the connection is easy.
2. The use of advanced digital filtering technology can effectively reduce measurement noise and improve measurement accuracy.
3. At the same time, we provide users with various PC terminals , instructions for use, and development manuals, so as to minimize the R&D time for various needs.
4. Supports 485 and CAN interfaces, allowing users to easily select the optimal connection method. The serial port rate is adjustable from 4800bps to 230400bps. The cutoff frequency is adjustable from 0 to 200Hz, and the detection period is adjustable from 1 to 200Hz.
5. Application areas: It can be widely used in bearing vibration measurement and real-time monitoring of rotating machinery such as submersible pumps, fans, steam turbines, coal mills, oxygen generators, generators, centrifuges, compressors, water pumps, motors, etc.
6. Three-axis displacement, three-axis velocity, and three-axis frequency outputs enable users to comprehensively measure vibration and shock, determining whether the measured object (motor or water pump) is damaged. If a machine malfunction is caused by factors such as bearing wear, cracking, poor dynamic balance, and misalignment, the vibration sensor can detect the fault in advance and issue an early warning, preventing the machine from continuing to operate under adverse conditions and causing damage, which could lead to economic losses.
7. Multiple installation methods: magnetic connection, threaded connection. Firm and stable, easy to install and disassemble.
8. Industrial-grade temperature sensing element, faster and more accurate acquisition of ambient temperature

2. Parameter indicators

2.1. Basic parameters

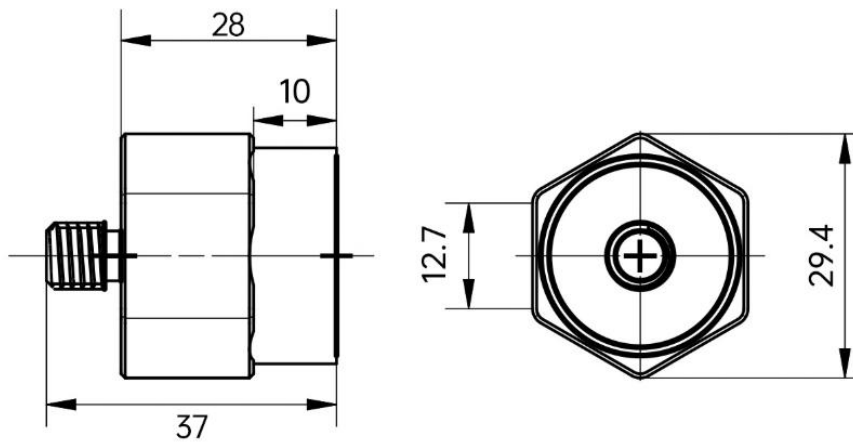
Parameter	Condition	Minimum	Default	Maximum
	485	4800bps	9600bps	230400bps

Communication interface	CAN	3K	250K	1M
Output content	On-chip time, 3-axis acceleration, 3-axis vibration velocity, 3-axis vibration displacement, 3-axis vibration frequency , temperature			
Range	Vibration speed: 0~100mm/s, vibration displacement: 0~30000um, vibration frequency: 5~2000Hz, temperature: -20℃~85℃			
Accuracy	<FS±4%			
Detection cycle		1Hz	10Hz	200Hz
Cutoff frequency		0Hz	10Hz	200Hz
Operating temperature		-20℃		85℃
Storage temperature		-40℃		85℃
Impact resistant				20000g
Protection level	IP67			

2.2. Parameter comparison

Product Appearance					
Model		WTVB02-485	WTVB01-485	WTVB05	WTVB01-BT50
Output	Vibration acceleration	✓	✓	✓	/
	Vibration speed	✓	✓	✓	✓
	Vibration displacement	✓	✓	✓	✓
	Vibration frequency	✓	✓	✓	✓
	Chip Time	✓	✓	✓	/
	Chip temperature	✓	✓	Ambient temperature	✓
Output rate	Normal mode (1-200Hz)	Send a command and return once(1-200Hz)			1-100Hz (default 20Hz)
Range	Vibration acceleration	±16G	±16G	±16G	/
	Vibration displacement	0~30000um	0-30000um	0-30000um	0-30000um
	Vibration speed	0~100mm/s	0-50mm/s	0-50mm/s	0-50mm/s
	Vibration frequency	5-2000Hz	5-100Hz	5-2000Hz	5-100Hz
Accuracy		<F.S±4%	<F.S±4%	<F.S±4%	<F.S±5%
Chip sampling rate		16KHz	1KHz	16KHz	1KHz
Communication interface		RS485	RS485	RS485/CAN	BLE5.0/TYPE-C
Communication Protocol		Modbus protocol	Modbus protocol	Modbus protocol CAN protocol	BLE5.0 protocol
Baud rate	RS485	4800bps-230400bps (default 9600bps)			115200bps (fixed)
	CAN	/	/	3K-1M(default 250K)	/
Temperature	Operating temperature	-40°C~85°C			-20°C~+60°C
	Storage temperature	-40°C~85°C			
Waterproof grade		IP67		Not waterproof	
Operating voltage		9-36V			5V
Working current		8mA		12mA (12V)	15mA
Weight		182g		67.5g	20g
Size		47*38*35mm			51.5*36.1*15mm

2.3. Product size



2.4. Electrical parameters

Parameter	Condition	Minimum	Default	Maximum
Supply voltage		5V	12V	36V
Working current	Working (12V)		12mA	

Vibration velocity: the speed at which the vibration point of an object moves when it vibrates

Amplitude: The amplitude of the vibration point when the object vibrates (vibration displacement)

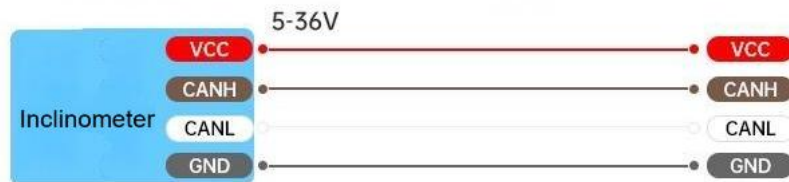
Vibration frequency: The number of times an object shakes per unit time during vibration. The vibration frequency measurement range is 1Hz~2000Hz

3. Hardware connection method

3.1. Wiring method

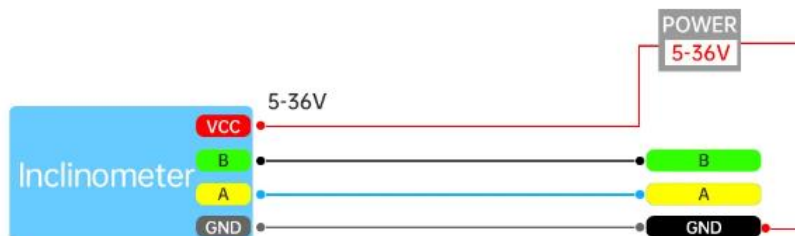
(CAN wiring method)

Writing	Red	Gray	Brown	White
	VCC	GND	CANH	CANL



(485 Wiring Method)

Wire Color Function	RED	YELLOW	GREEN	BLACK
	VCC 5~36V	A	B	GND



3.2. Installation

Threaded/Magnetic Mounting More Convenient!

Two flexible installation options to meet diverse application needs. You can freely choose between flexible mounting or adhesive fixation.



Threaded Mounting
Thread (M8, M5)



Magnetic Mounting
Magnetic base for easy
installation and removal.

4. How to use the software

4.1. PC software download

1. CH340 driver download link:

https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

2. Witmotion software download link:

https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

4.2. Connect to the PC software

to the computer via a USB adapter , open Witmotion.exe , and after installing the CH340 driver, you can query the corresponding port number in the device manager, as shown in the figure:

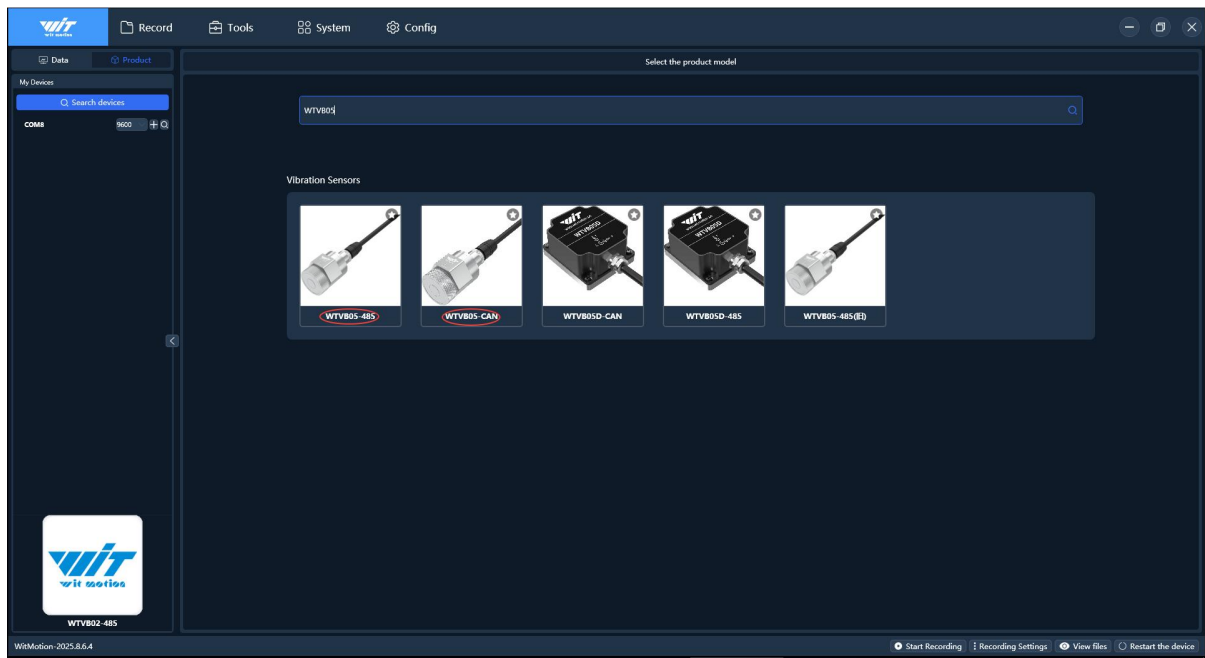


After the device is connected, open the Witmotion.exe software.

名称	修改日期	类型	大小
AutoUpdateApp	2024/9/6 星期五 17:21	文件夹	
Bin	2025/4/8 星期二 12:05	文件夹	
Config	2025/4/29 星期二 14:...	文件夹	
Plugins	2024/9/6 星期五 17:21	文件夹	
Record	2025/4/29 星期二 14:...	文件夹	
Temp	2024/9/6 星期五 17:24	文件夹	
1.txt	2025/4/18 星期五 19:...	文本文档	1 KB
debug.log	2025/4/18 星期五 19:...	文本文档	1 KB
WitMotion.exe	2025/4/16 星期三 17:...	应用程序	1,543 KB
WitMotion.exe.config	2025/4/16 星期三 17:...	CONFIG 文件	19 KB
WitMotion.pdb	2025/4/16 星期三 17:...	PDB 文件	424 KB

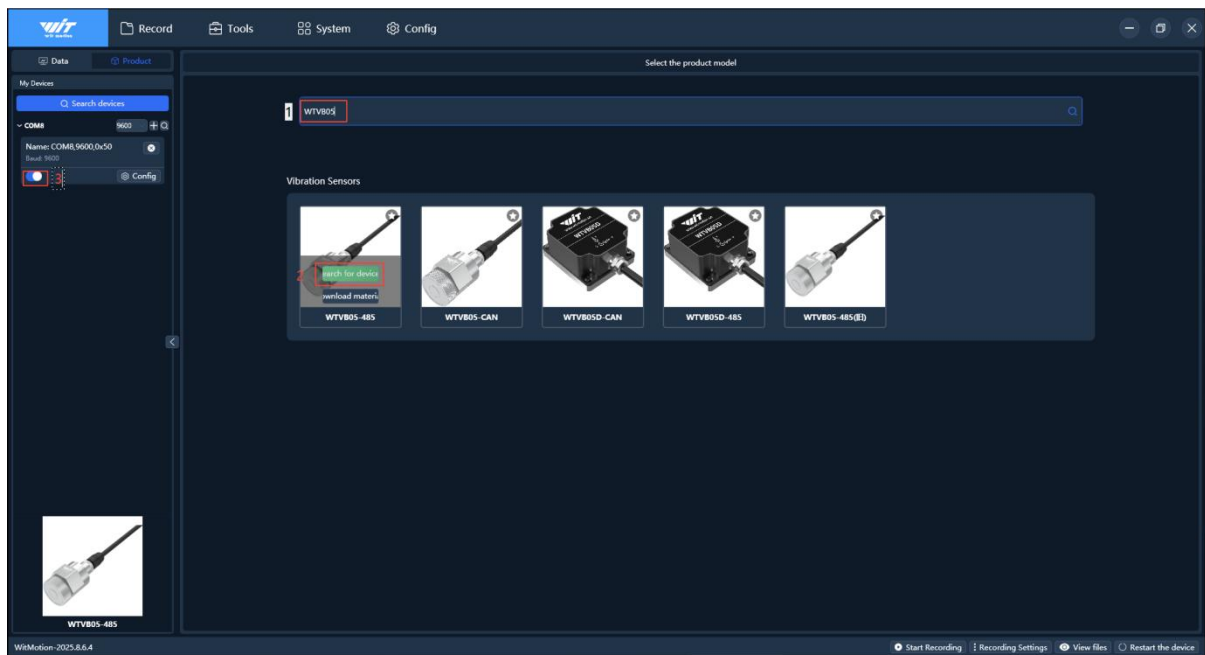
4.2.1. Automatic Search

1. Select WTVB05-485/WTVB05-CAN (WTVB05-485 is used for demonstration below).

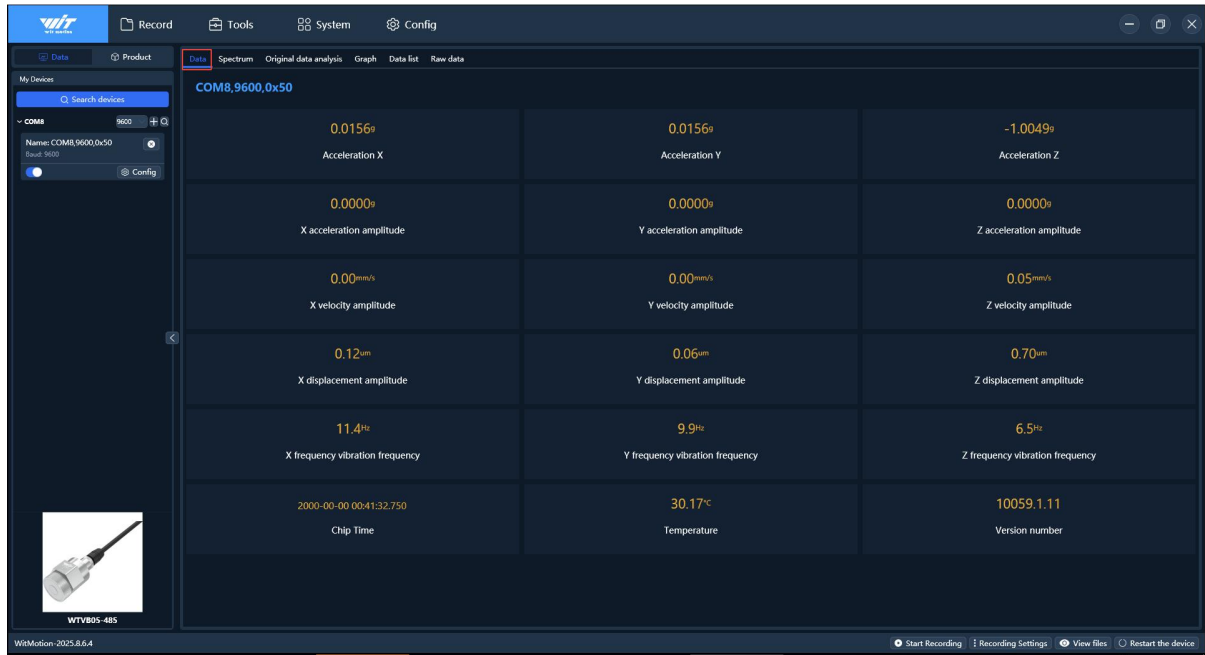


2. Click Search device.

3. Check the serial port number.



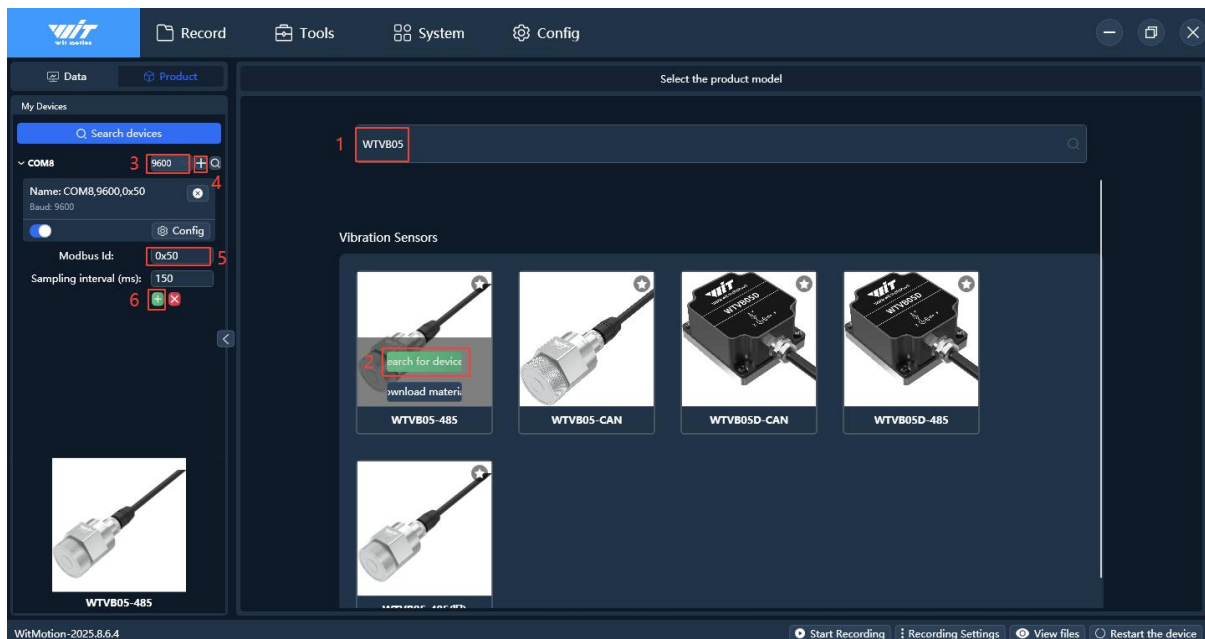
After the connection is successful, the data will be displayed on the software, as shown below:



4.2.2. Manual connection

1. Select the model number WTVB05-485.
2. Select the baud rate of the corresponding port. The default baud rate is 9600
3. Enter the corresponding ID (the default is 0x50) and click the + sign to add the device.

As shown in the figure below:



4.3. Graph

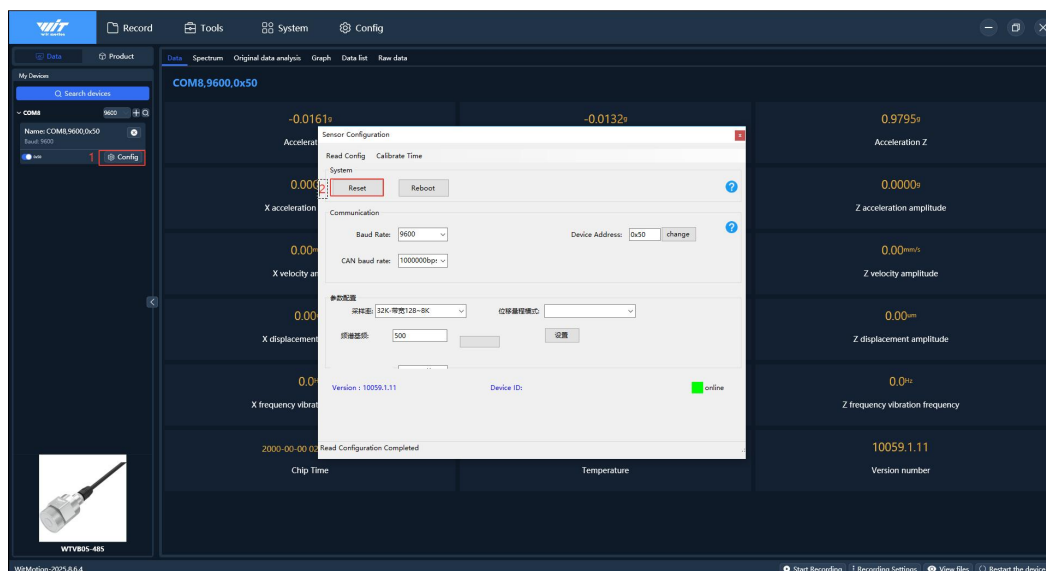
Click on the graph to see graphs of vibration speed, vibration frequency, vibration displacement and temperature:



5. Software Configuration

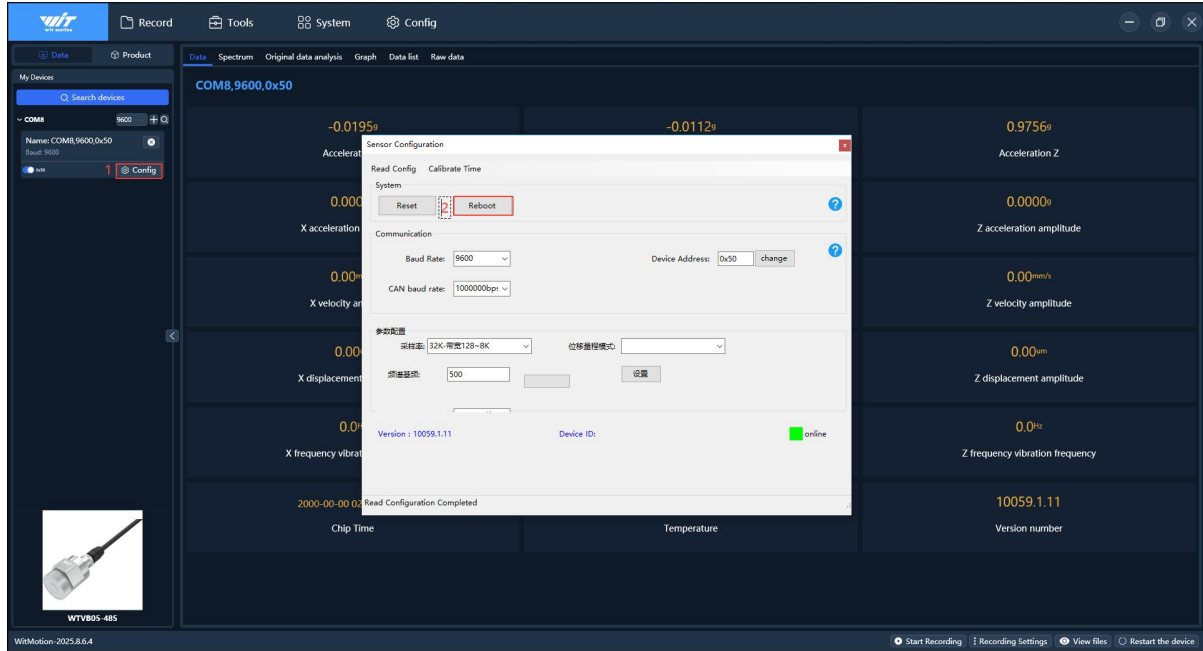
5.1. Reset Settings

Click "Configuration" and then click "Reset Settings" in the sensor configuration interface to Reset the factory settings, as shown below:



5.2. Reboot

Click "Config" and then click "Reboot" in the sensor configuration interface to Reboot the sensor, as shown below:



5.3. Communication rate

Open "Config" and click the "Baud Rate" drop-down menu in the sensor configuration interface. Select the serial port baud rate to be modified to change the current serial port baud rate (the default serial port baud rate is 9600). The serial port baud rate can be 4800, 9600, 19200, 38400, 57600, 115200, and 230400, as shown in the figure below:

Read Config

Calibrate Time

System

Reset

Reboot

Communication

Baud Rate:

9600

4800

9600

19200

38400

57600

115200

230400

CAN baud rate:

32K-带宽128~8K

Device Address:

0x50

change

参数配置

采样率:

32K-带宽128~8K

位移量程模式:

频谱基频:

500

设置

Version : 10059.1.11

Device ID:

online

Read Configuration Completed

5.4. Device Address

Open "Configuration" and click the "Modbus Address" input box in the sensor configuration interface. Enter the Modbus address and click Set to change the Modbus address (the default Modbus address is 0x50). The Modbus address ranges from 0x00 to 0x7F, as shown in the figure:

Read Config
Calibrate Time

System

Reset
Reboot

Communication

Baud Rate: 9600
Device Address: 0x50 change

CAN baud rate: 1000000bp

参数配置

采样率: 32K-带宽128~8K
位移量程模式:

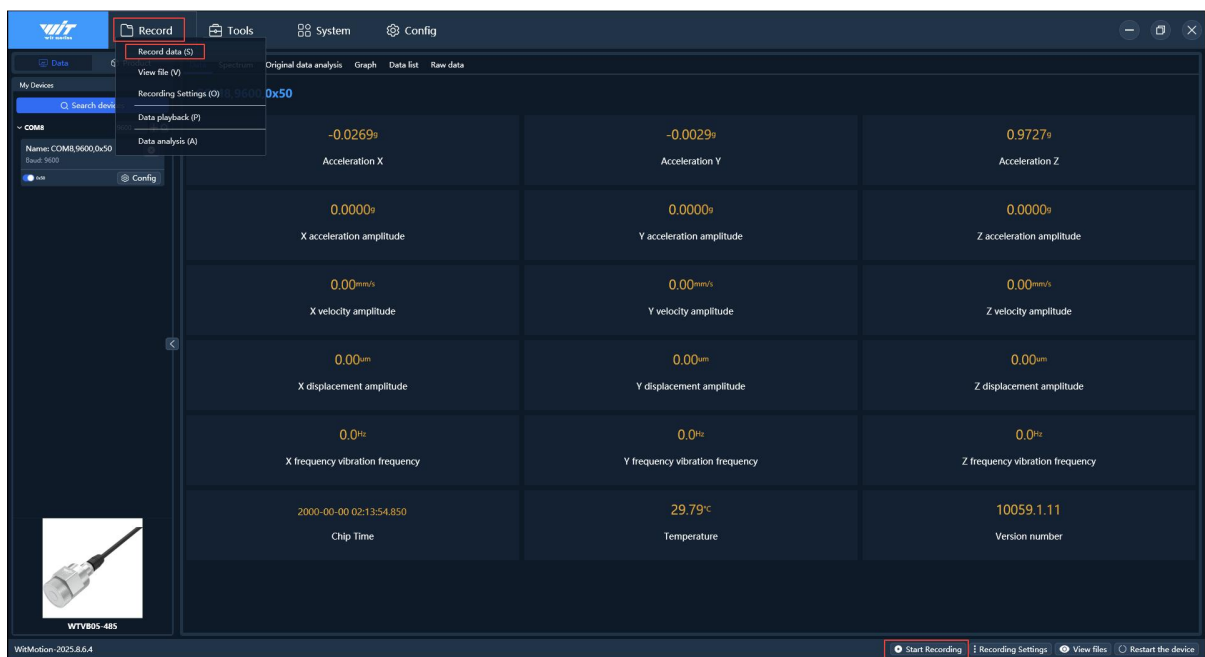
频谱基频: 500
设置

Version : 10059.1.11
Device ID:
online

Setting succeeded!

5.5. Recording Data

Open "Record" and click "Start Recording" to record the output data of the sensor.



Record data (3)	Original data analysis	Graph	Data list	Raw data	
Record data (3)	View file (V)				
Recording Settings (O)					
Data playback (P)					
Data analysis (A)					
Acceleration X	-0.0269g	Acceleration Y	-0.0029g	Acceleration Z	0.9727g
X acceleration amplitude	0.0000g	Y acceleration amplitude	0.0000g	Z acceleration amplitude	0.0000g
X velocity amplitude	0.00mm/s	Y velocity amplitude	0.00mm/s	Z velocity amplitude	0.00mm/s
X displacement amplitude	0.00mm	Y displacement amplitude	0.00mm	Z displacement amplitude	0.00mm
X frequency vibration frequency	0.0Hz	Y frequency vibration frequency	0.0Hz	Z frequency vibration frequency	0.0Hz
Chip Time	2000-00-00 02:13:54.850	Temperature	29.79°C	Version number	10059.1.11

Start Recording
Recording Settings
View files
Restart the device

6. Communication Protocol

6.1. MODBUS protocol

6.1.1. Read register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address, register count, and data read are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, the high and low bits of the register count are represented by LEN_H and LEN_L, and the high and low bits of the data read are represented by DATA_{1H} and DATA_{1L}.
- The last two digits of the read instruction are standard CRC check digits. You can use a CRC check digit calculation tool.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	Read the high 8 bits of the length	Read length lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	ADDR _H [15:8]	ADDR _L [7:0]	LEN _H [3:08 PM]	LEN _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Read length	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	LEN[7:0]	DATA _{1H} [15:8]	DATA _{1L} [7:0]	...	...	DATA _{nH}	DATA _{nL}	CRCH[15:8]	CRCL[7:0]

6.1.2. Write register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address and write data are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, and the high and low bits of the write data are represented by DATA_H and DATA_L.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Note:

The instruction writing operation needs to be performed in three steps.

The first step is to unlock 0x50 0x06 0x00 0x69 0xB5 0x88 0x22 0xA1. The unlocking will take effect within ten seconds.

The second step is to send the instructions that need to be modified.

The third step is to save the instruction 0x50 0X06 0X00 0X00 0X00 0X84 0X4B.

6.1.3. Register Address Table

Register address	symbol	meaning
0x00	SAVE	Save/Reboot/Reset to Factory
0x03	RRATE	Return rate
0x04	BAUD	Serial port baud rate
0x05	CANBAUD	CAN baud rate
0x07	MODE	Algorithm Mode
0x08	STATIC DETECTION	Static detection coefficient
0x1A	IICADDR	Device Address

0x30	YYMM	Month Year
0x31	DDHH	Time
0x32	MMSS	Seconds and minutes
0x33	MS	millisecond
0x34	AX	X-axis acceleration
0x35	AY	Y-axis acceleration
0x36	AZ	Z- axis acceleration
0x3A	VX	X-axis vibration speed
0x3B	VY	Y-axis vibration speed
0x3C	VZ	Z-axis vibration speed
0x40	TEMP	Product temperature
0x41	DX	X-axis vibration displacement
0x42	DY	Y-axis vibration displacement
0x43	DZ	Z-axis vibration displacement
0x44	HZX	X-axis vibration frequency
0x45	HZY	Y-axis vibration frequency
0x46	HZZ	Z-axis vibration frequency

6.1.4. Register Description

All the following examples are instructions for when the Modbus address is 0x50 (default). If you change the Modbus address, you need to change the address and CRC check bit in the instruction accordingly.

6.1.4.1. SAVE (Save/Reboot/Reset to Factory)

Register Name: SAVE

Register address: 0 (0x00)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Reboot: 0x00FF Factory Reset: 0x0001

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 00 00 FF C4 0B (Reboot)

Return: 50 06 00 00 00 FF C4 0B

6.1.4.2. BAUD (serial port baud rate)

Register Name: BAUD

Register address: 4 (0x04)

Read/write direction: R/W

Default value: 0x0002

Bit	NAME	FUNCTION
15:4		
3:0	BAUD[3:0]	Set the serial port baud rate: 0001(0x01) 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps

		0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps
Example: Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking) Return: 50 06 00 69 B5 88 22 A1 Send: 50 06 00 04 00 06 45 88 (set the serial port baud rate to 115200) Send: 50 06 00 00 00 00 84 4B (saved at 115200 baud) Return: 50 06 00 00 00 00 84 4B		

6.1.4.3. CANBAUD (CAN baud rate)

Register Name: CANBAUD

Register address: 5 (0x05)

Read/write direction: R/W

Default value: 0x0004

Bit	NAME	FUNCTION
15:4		
3:0	CANBAUD[3:0]	Set the CAN baud rate: 0000(0x00): 1000000bps 0001(0x01): 800000bps 0010(0x02): 500000bps 0011(0x03): 400000bps 0100(0x04): 250000bps 0101(0x05): 200000bps 0110(0x06): 125000bps 0111(0x07): 100000bps

		1000(0x08): 80000bps 1001(0x09): 50000bps 1010(0x0A): 40000bps 1011(0x0B): 20000bps 1100(0x0C): 10000bps 1101(0x0D): 5000bps 1110(0x0E): 3000bps
Example: Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking) Return: 50 06 00 69 B5 88 22 A1 Send: 50 06 00 05 00 02 15 8B (set CAN baud rate to 500K) Send: 50 06 00 00 00 00 84 4B (saved at 500Kbaud) Return: 50 06 00 00 00 00 84 4B		

6.1.4.4. MODE (algorithm mode)

Register Name: MODE		
Register address: 7 (0x07)		
Read/write direction: R/W		
Default value: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	MODE[1:0]	Set the algorithm mode: 0000(0x00): Time domain low frequency algorithm(10~200Hz) 0001(0x01): Time domain high frequency algorithm 0010(0x02): Frequency domain algorithm

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 07 00 01 F4 4A (set time domain high frequency algorithm)

Send: 50 06 00 00 00 00 84 4B

Return: 50 06 00 00 00 00 84 4B

6.1.4.5. STATICDETECTION (static detection coefficient)

Meaning: After the device is stationary, there is still vibration data. The speed at which the data is cleared has no unit meaning and is a coefficient.

Register Name: STATICDETECTION

Register address: 8 (0x08)

Read/write direction: R/W

Default value: 0x0064

Bit	NAME	FUNCTION
15:4		
3:0	MODE[3:0]	Set the static detection coefficient: Range: 80~120 $100 / 100 = 1.00$, default coefficient 1.00

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 08 00 65 C5 A2 (set static detection coefficient to 1.01, write 101)

Send: 50 06 00 00 00 00 84 4B

Return: 50 06 00 00 00 00 84 4B

6.1.4.6. IICADDR (device address)

Register Name: IICADDR

Register Address: 26 (0x1A)

Read/write direction: R/W

Default value: 0x0050

Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address, used for I2C and Modbus communication, use 0x01~0x7F

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 1A **00 02** 24 4D (set device address to **0x02**)

Return: 50 06 00 1A 00 02 24 4D

Send: **02** 06 00 69 B5 88 2F 13 (valid within 10 seconds after unlocking)

Return: 02 06 00 69 B5 88 2F 13

Send: **02** 06 00 00 00 00 89 F9 (Save)

Return: 02 06 00 00 00 00 89 F9

6.1.4.7. YYMM~MS (on-chip time)

Register name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Read/write direction: R/W

Default time: (2015, 1, 1, 00, 00, 59, 00)

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	moon
7:0	YYMM[7:0]	Year
15:8	DDHH[15:8]	hour
7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	Second
7:0	MMSS[7:0]	point
15:0	MS[15:0]	millisecond

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 30 03 16 05 7A (set year and month 22-03)

Return: 50 06 00 30 03 16 05 7A

Send: 50 06 00 31 09 0C D3 D1 (set date and time 12-09)

Return: 50 06 00 31 09 0C D3 D1

Send: 50 06 00 32 3A 1E B7 2C (set minutes and seconds to 30:58)

Return: 50 06 00 32 3A 1E B7 2C

Send: 50 06 00 33 01 F4 74 53 (set milliseconds to 500)

Return: 50 06 00 33 01 F4 74 53

Send: 50 06 00 00 00 00 84 4B (save)

Return: 50 06 00 00 00 00 84 4B

6.1.4.8. AX~AZ (acceleration)

Register name: **AX~AZ**

Register address: 52~54 (0x34~0x36)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X = AX[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s²)
15:0	AY[15:0]	Acceleration Y = AY[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s²)
15:0	AZ[15:0]	Acceleration Z = AZ[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s²)

Example:

Send: 50 03 00 34 00 03 49 84 (read three-axis acceleration)

Return: 50 03 06 AXH AXL AYH AYL AZH AZL CRCH CRCL

AX[15:0]=((short)AXH <<8)|AXL;

AY[15:0]=((short)AYH <<8)|AYL;

AZ[15:0]=((short)AZH <<8)|AZL;

6.1.4.9. VX~VZ (vibration speed)

10141.1.7 (including 1.7) and earlier versions, the vibration speed does not need to be divided by 100

Register name: VX~VZ

Register address: 58~60 (0x3A~0x3C)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
-----	------	----------

15:0	VX[15:0]	Vibration speed VX (mm/s) = ((VXH << 8) VXL)/100
15:0	VY[15:0]	Vibration velocity VY (mm/S) = ((VYH << 8) VYL)/100
15:0	VZ[15:0]	Vibration velocity VZ (mm/s) = ((VZH << 8) VZL)/100

Example:

Send: 50 03 00 3A 00 03 28 47 (read the three-axis vibration speed)

Return: 50 03 06 VXH VXL VYH VYL VZH VZL CRCH CRCL

VX[15:0]=(((short)VXH <<8)|VXL)/100;

VY[15:0]=(((short)VYH <<8)|VYL)/100;

VZ[15:0]=(((short)VZH <<8)|VZL)/100;

6.1.4.10. Reserve

Register Name: Reserved

Register address: 61~63 (0x3D~0x3F)

Read/write direction: R

Default value: 0x0000

6.1.4.11. TEMP (temperature)

Register Name: TEMP

Register address: 64 (0x40)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature = TEMP[15:0]/100°C

Example:

Send: 50 03 00 40 00 01 88 5F (read chip temperature)

Return: 50 03 02 TEMPH TEMPL CRCH CRCL
 TEMP[15:0]=(((short)TEMPH <<8)|TEMPL);

6.1.4.12. DX~DZ (vibration displacement)

Register name: DX~DZ

Register address: 65~67 (0x41~0x43)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	DX[15:0]	Vibration displacement DX (um) = ((DXH << 8) DXL)
15:0	DY[15:0]	Vibration displacement DY (um) = ((DYH << 8) DYL)
15:0	DZ[15:0]	Vibration displacement DZ (um) = ((DZH << 8) DZL)

Example:

Send: 50 03 00 41 00 03 58 5E (read triaxial vibration displacement)

Return: 50 03 06 DXH DXL DYH DYL DZH DZL CRCH CRCL

DX[15:0]=(((short)DXH <<8)|DXL);

DY[15:0]=(((short)DYH <<8)|DYL);

DZ[15:0]=(((short)DZH <<8)|DZL);

6.1.4.13. HZX~HZZ (vibration frequency)

Register name: HZX~HZZ

Register address: 68~70 (0x44~0x46)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
-----	------	----------

15:0	HZX[15:0]	Vibration frequency HZX(Hz)=((HZZH << 8) HZZL)/10
15:0	HZY[15:0]	Vibration frequency HZY(Hz)=((HZZH << 8) HZZL)/10
15:0	HZZ[15:0]	Vibration frequency HZZ (Hz) = ((HZZH << 8 HZZL)/10

Example:

Send: 50 03 00 44 00 03 48 5F (read the three-axis vibration frequency)

Return: 50 03 06 HZXH HZXL HZYH HZYL HZZH HZZL CRCH CRCL

$HZX[15:0]=(((\text{short})HZZH \ll 8)| HZZL)/10;$

$HZY[15:0]=(((\text{short})HZZH \ll 8)| HZZL)/10;$

$HZZ[15:0]=(((\text{short})HZZH \ll 8)| HZZL)/10;$

6.1.4.14. Displacement range mode

Register name: DRM

Register address: 43 (0x2B)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	DRM[1:0]	Set displacement range mode: 0001(0x01): 60000um-1um resolution 0010(0x02): 600um-0.01um resolution

Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 2B 00 02 75 82 (Sets resolution from 600um to 0.01um)

Return: 50 06 00 2B 00 02 75 82

Send: 50 06 00 00 00 00 84 4B

Return: 50 06 00 00 00 00 84 4B

6.1.4.15. Sampling rate

Register name: SR

Register address: 41 (0x29)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	SR[1:0]	Set sampling rate: 0000(0x00): 32K - Measurable frequency 128~16KHz 0001(0x01): 16K - Measurable frequency 64~8KHz 0010(0x02): 8K - Measurable frequency 32~4KHz 0011(0x03): 4K - Measurable frequency 16~2KHz 0100(0x04): 2K - Measurable frequency 8~1KHz 0101(0x05): 1K - Measurable frequency 4~512Hz 0110(0x06): 512Hz - Measurable frequency 2~256Hz 0111(0x07): 256Hz - Measurable frequency 1~128Hz 1000(0x08): 128Hz - Measurable frequency 1~64Hz 1001(0x09): 64Hz - Measurable frequencies 0.1~32Hz

Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 29 00 07 14 41 (Setting to 256Hz - Measurable frequency 1~128Hz)

Return: 50 06 00 29 00 07 14 41

Send: 50 06 00 00 00 00 84 4B

Return: 50 06 00 00 00 00 84 4B

6.1.4.16. Original Acceleration FIFO

Register name: RAWFIFO

Register address: 414(0x2C)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	RAWFIFO[1:0]	

Example:

Send: 50 03 00 2C 00 01 48 42

Return: Returns raw accelerometer data, buffering 1024 packets.

Note: After sending the command to read the raw acceleration FIFO, the sensor first reads 1024 data entries.

The acceleration XYZ data consists of 6 bytes in total, with a data size of 6144 bytes (excluding header and checksum).

If the sensor has not yet read 1024 data entries, it will return 50 03 01 XX XX CRCH CRCL, where XX XX represents the current data size.

If the sensor has already read 1024 data entries, it will return 50 03 00 AXH AXL AYH AYL AZH AZL AXH AXL AYH AYL

AZH AZL CRCH CRCL. The acceleration XYZ data is converted to g, as described in section **6.4.1.6**, acceleration data calculation.

6.1.4.17. X-axis vibration simulation quantity

Register name: CFX~DRMSX

Register address: 71~82 (0x47~0x52)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFX[15:0]	X-axis acceleration peak factor $CFX(g) = ((CFXH \ll 8) \mid CFXL)/1000$
15:0	KX[15:0]	X-axis acceleration kurtosis value $KX(g) = ((KXH \ll 8) \mid KXL)/1000$
15:0	AAVGX[15:0]	The absolute average value of the X-axis acceleration, $AAVGX(g)$, is calculated as follows: $AAVGXH \ll 8 \mid AAVGXL)/1000$
15:0	VARX[15:0]	variance of X-axis acceleration, $VARX(g)$, is calculated as: $VARXH \ll 8 \mid VARXL)/1000$.
15:0	RRAX[15:0]	X-axis acceleration root mean (RMS) $RRAX(g) = ((RRAXH \ll 8 \mid RRAXL)/1000)$
15:0	WIX[15:0]	X-axis acceleration waveform index $WIX = ((WIXH \ll 8 \mid WIXL)/1000)$
15:0	PIX[15:0]	X-axis acceleration peak index $PIX(g) = ((PIXH \ll 8 \mid PIXL)/1000)$
15:0	PCX[15:0]	X-axis acceleration impulse coefficient $PCX = ((PCXH \ll 8 \mid PCXL)/1000)$
15:0	SKX[15:0]	X-axis acceleration skew $SKX(g) = ((SKXH \ll 8 \mid SKXL)/1000)$
15:0	VRMSX[15:0]	X-axis velocity RMS value $VRMSX(mm/s) = ((VRMSXH \ll 8 \mid VRMSXL)/1000)$
15:0	VKX[15:0]	X-axis velocity kurtosis value $VKX(g) = ((VKXH \ll 8 \mid VKXL) / 1000)$
15:0	DRMSX[15:0]	X-axis displacement RMS value $DRMSX(mm) = ((DRMSXH \ll 8 \mid DRMSXL) / 1000)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 47 00 01 39 9E (X-axis acceleration peak factor)

Return: 50 03 06 CFXH CFXL CRCH CRCL

CFX[15:0]=(((short)CFXH <<8)|CFXL)/1000;

6.1.4.18. Y-axis vibration simulation quantity

Register name: CFY~DRMSY

Register address: 83~94 (0x53~0x5E)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFY[15:0]	Y-axis acceleration peak factor CFY(g) = ((CFYH << 8) CFYL)/1000
15:0	KY[15:0]	Y-axis acceleration kurtosis value KY(g) = ((KYH << 8) KYL)/1000
15:0	AAVGY[15:0]	The absolute average value of the Y-axis acceleration, AAVGY(g), is calculated as follows: AAVGYH << 8 AAVGYL)/1000
15:0	VARY[15:0]	variance of Y-axis acceleration, VARY(g), is calculated as: VARYH << 8 VARYL)/1000.
15:0	RRAY[15:0]	Y-axis acceleration root mean (RMS) RRAY(g) = ((RRAYH << 8 RRAYL)/1000)
15:0	WIY[15:0]	Y-axis acceleration waveform index WIY = ((WIYH << 8 WIYL)/1000)
15:0	PIY[15:0]	Y-axis acceleration peak index PIY(g) = ((PIYH << 8 PIYL)/1000)
15:0	PCY[15:0]	Y-axis acceleration impulse coefficient PCY = ((PCYH << 8 PCYL)/1000)
15:0	SKY[15:0]	Y-axis acceleration skew SKY(g) = ((SKYH << 8 SKYL)/1000)

15:0	VRMSY[15:0]	Y-axis velocity RMS value VRMSY(mm/s)=((VRMSYH<< 8 VRMSYL)/1000
15:0	VKY[15:0]	Y-axis velocity kurtosis value VKY (g) = ((VKYH << 8 VKYL) / 1000
15:0	DRMSY[15:0]	Y-axis displacement RMS value DRMSY (mm) = ((DRMSYH << 8 DRMSYL) / 1000

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 47 00 01 39 9E (Y-axis acceleration peak factor)

Return: 50 03 06 CFYH CFYL CRCH CRCL

CFY[15:0]=(((short)CFYH <<8)|CFYL)/1000;

6.1.4.19. Z-axis vibration simulation quantity

Register name: CFZ~DRMSZ		
Register address: 95~106 (0x5F~0x6A)		
Read/write direction: R		
Default value: 0x0000		
Bit	NAME	FUNCTION
15:0	CFZ[15:0]	Z-axis acceleration peak factor CFZ(g) = ((CFZH << 8) CFZL)/1000
15:0	KZ[15:0]	Z-axis acceleration kurtosis value KZ(g) = ((KZH << 8) KZL)/1000
15:0	AAVGZ[15:0]	The absolute average value of the Z-axis acceleration, AAVGZ(g), is calculated as follows: AAVGZH << 8 AAVGZL)/1000
15:0	VARZ[15:0]	variance of Z-axis acceleration, VARZ(g), is calculated as: VARZH << 8 VARZL)/1000.
15:0	RRAZ[15:0]	Z-axis acceleration root mean (RMS) RRAZ(g) = ((RRAZH << 8 RRAZL)/1000)

15:0	WIZ[15:0]	Z-axis acceleration waveform index $WIZ = ((WIZH < 8 \mid WIZL)/1000)$
15:0	PIZ[15:0]	Z-axis acceleration peak index $PIZ(g) = ((PIZH < 8 \mid PIZL)/1000)$
15:0	PCZ[15:0]	Z-axis acceleration impulse coefficient $PCZ = ((PCZH < 8 \mid PCZL)/1000)$
15:0	SKZ[15:0]	Z-axis acceleration skew $SKZ(g) = ((SKZH < 8 \mid SKZL)/1000)$
15:0	VRMSZ[15:0]	Z-axis velocity RMS value $VRMSZ(mm/s) = ((VRMSZH < 8 \mid VRMSZL)/1000)$
15:0	VKZ[15:0]	Z-axis velocity kurtosis value $VKZ(g) = ((VKZH < 8 \mid VKZL) / 1000)$
15:0	DRMSZ[15:0]	Z-axis displacement RMS value $DRMSZ(mm) = ((DRMSZH < 8 \mid DRMSZL) / 1000)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

$NAME[15:0] = (((short)NAMEH < 8) \mid NAMEL)/1000;$

Example:

Send: 50 03 00 47 00 01 39 9E (Z-axis acceleration peak factor)

Return: 50 03 06 CFZH CFZL CRCH CRCL

$CFZ[15:0] = (((short)CFZH < 8) \mid CFZL)/1000;$

6.1.4.20. XYZ axis fault diagnosis

Register name: ERRXERRZ

Register address: 107~109 (0x6B~0x6D)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	ERRX	
15:0	ERRY	

15:0	ERRZ	
------	------	--

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6B 00 01 F8 57

Return: 50 03 06 ERRXH ERRXL CRCH CRCL

ERRX[15:0]=(((short)ERRXH <<8)|ERRXL);

6.1.4.21. X-axis focus and acc spectrum energy

Register name: AX_ENERGY_POINT1~AX_ENERGY

Register address: 110~124 (0x6E~0x7C)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX_ENERGY_POINT1[15:0]	X-axis focus point 1: Acceleration spectrum energy AX_ENERGY_POINT1 = ((AX_ENERGY_POINT1H << 8) AX_ENERGY_POINT1L)
15:0	AX_ENERGY_POINT2[15:0]	X-axis focus point 2: Acceleration spectrum energy AX_ENERGY_POINT2 = ((AX_ENERGY_POINT2H << 8) AX_ENERGY_POINT2L)
15:0	AX_ENERGY_POINT3[15:0]	X-axis focus point 3: Acceleration spectrum energy AX_ENERGY_POINT3 = ((AX_ENERGY_POINT3H << 8) AX_ENERGY_POINT3L)
15:0	AX_ENERGY_POINT4[15:0]	X-axis focus point 4: Acceleration spectrum energy AX_ENERGY_POINT4 = ((AX_ENERGY_POINT4H << 8) AX_ENERGY_POINT4L)
15:0	AX_ENERGY_POINT5[15:0]	X-axis focus point 5: Acceleration spectrum energy AX_ENERGY_POINT5 = ((AX_ENERGY_POINT5H << 8) AX_ENERGY_POINT5L)
15:0	AX_ENERGY_POINT6[15:0]	X-axis focus point 6: Acceleration spectrum energy AX_ENERGY_POINT6 = ((AX_ENERGY_POINT6H << 8) AX_ENERGY_POINT6L)

15:0	AX_ENERGY_POINT7[15:0]	X-axis focus point 7: Acceleration spectrum energy AX_ENERGY_POINT7 = ((AX_ENERGY_POINT7H << 8) AX_ENERGY_POINT7L)
15:0	AX_ENERGY_POINT8[15:0]	X-axis focus point 8: Acceleration spectrum energy AX_ENERGY_POINT8 = ((AX_ENERGY_POINT8H << 8) AX_ENERGY_POINT8L)
15:0	AX_ENERGY_AREA1[15:0]	X-axis focus band 1 acceleration spectrum energy AX_ENERGY_AREA1=((AX_ENERGY_AREA1H<< 8 AX_ENERGY_AREA1L)
15:0	AX_ENERGY_AREA2[15:0]	X-axis focus band 2 acceleration spectrum energy AX_ENERGY_AREA2=((AX_ENERGY_AREA2H<< 8 AX_ENERGY_AREA2L)
15:0	AX_ENERGY_AREA3[15:0]	X-axis focus band 3 acceleration spectrum energy AX_ENERGY_AREA3=((AX_ENERGY_AREA3H<< 8 AX_ENERGY_AREA3L)
15:0	AX_ENERGY_AREA4[15:0]	X-axis focus band 4 acceleration spectrum energy AX_ENERGY_AREA4=((AX_ENERGY_AREA4H<< 8 AX_ENERGY_AREA4L)
15:0	AX_ENERGY_AREA5[15:0]	X-axis focus band 5 acceleration spectrum energy AX_ENERGY_AREA5=((AX_ENERGY_AREA5H<< 8 AX_ENERGY_AREA5L)
15:0	AX_ENERGY_AREA6[15:0]	X-axis focus band 6 acceleration spectrum energy AX_ENERGY_AREA6=((AX_ENERGY_AREA6H<< 8 AX_ENERGY_AREA6L)
15:0	AX_ENERGY[15:0]	Total energy of the X-axis acceleration spectrum AX_ENERGY = ((AX_ENERGYH << 8 AX_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (X-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AX_ENERGY_POINT1H AX_ENERGY_POINT1L CRCH CRCL

AX_ENERGY_POINT1[15:0]=(((short)AX_ENERGY_POINT1H <<8)|AX_ENERGY_POINT1L);

6.1.4.22. Y-axis focus and acc spectrum energy

Register name: AY_ENERGY_POINT1~AY_ENERGY

Register address: 125~139 (0x7D~0x8B)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AY_ENERGY_POINT1[15:0]	Y-axis focus point 1: Acceleration spectrum energy AY_ENERGY_POINT1 = ((AY_ENERGY_POINT1H << 8) AY_ENERGY_POINT1L)
15:0	AY_ENERGY_POINT2[15:0]	Y-axis focus point 2: Acceleration spectrum energy AY_ENERGY_POINT2 = ((AY_ENERGY_POINT2H << 8) AY_ENERGY_POINT2L)
15:0	AY_ENERGY_POINT3[15:0]	Y-axis focus point 3: Acceleration spectrum energy AY_ENERGY_POINT3 = ((AY_ENERGY_POINT3H << 8) AY_ENERGY_POINT3L)
15:0	AY_ENERGY_POINT4[15:0]	Y-axis focus point 4: Acceleration spectrum energy AY_ENERGY_POINT4 = ((AY_ENERGY_POINT4H << 8) AY_ENERGY_POINT4L)
15:0	AY_ENERGY_POINT5[15:0]	Y-axis focus point 5: Acceleration spectrum energy AY_ENERGY_POINT5 = ((AY_ENERGY_POINT5H << 8) AY_ENERGY_POINT5L)
15:0	AY_ENERGY_POINT6[15:0]	Y-axis focus point 6: Acceleration spectrum energy AY_ENERGY_POINT6 = ((AY_ENERGY_POINT6H << 8) AY_ENERGY_POINT6L)
15:0	AY_ENERGY_POINT7[15:0]	Y-axis focus point 7: Acceleration spectrum energy AY_ENERGY_POINT7 = ((AY_ENERGY_POINT7H << 8) AY_ENERGY_POINT7L)
15:0	AY_ENERGY_POINT8[15:0]	Y-axis focus point 8: Acceleration spectrum energy AY_ENERGY_POINT8 = ((AY_ENERGY_POINT8H << 8) AY_ENERGY_POINT8L)
15:0	AY_ENERGY_AREA1[15:0]	Y-axis focus band 1 acceleration spectrum energy AY_ENERGY_AREA1=((AY_ENERGY_AREA1H<< 8 AY_ENERGY_AREA1L)
15:0	AY_ENERGY_AREA2[15:0]	Y-axis focus band 2 acceleration spectrum energy AY_ENERGY_AREA2=((AY_ENERGY_AREA2H<< 8 AY_ENERGY_AREA2L)
15:0	AY_ENERGY_AREA3[15:0]	Y-axis focus band 3 acceleration spectrum energy AY_ENERGY_AREA3=((AY_ENERGY_AREA3H<< 8 AY_ENERGY_AREA3L)
15:0	AY_ENERGY_AREA4[15:0]	Y-axis focus band 4 acceleration spectrum energy AY_ENERGY_AREA4=((AY_ENERGY_AREA4H<< 8 AY_ENERGY_AREA4L)
15:0	AY_ENERGY_AREA5[15:0]	Y-axis focus band 5 acceleration spectrum energy AY_ENERGY_AREA5=((AY_ENERGY_AREA5H<< 8 AY_ENERGY_AREA5L)
15:0	AY_ENERGY_AREA6[15:0]	Y-axis focus band 6 acceleration spectrum energy AY_ENERGY_AREA6=((AY_ENERGY_AREA6H<< 8 AY_ENERGY_AREA6L)
15:0	AY_ENERGY[15:0]	Total energy of the Y-axis acceleration spectrum AY_ENERGY = ((AY_ENERGYH << 8 AY_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (Y-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AY_ENERGY_POINT1H AY_ENERGY_POINT1L CRCH CRCL

AY_ENERGY_POINT1[15:0]=(((short)AY_ENERGY_POINT1H <<8)|AY_ENERGY_POINT1L);

6.1.4.23. Z-axis focus and acc spectrum energy

Register name: AZ_ENERGY_POINT1~AZ_ENERGY

Register address: 140~154 (0x8C~0x9A)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AZ_ENERGY_POINT1[15:0]	Z-axis focus point 1: Acceleration spectrum energy AZ_ENERGY_POINT1 = ((AZ_ENERGY_POINT1H << 8) AZ_ENERGY_POINT1L)
15:0	AZ_ENERGY_POINT2[15:0]	Z-axis focus point 2: Acceleration spectrum energy AZ_ENERGY_POINT2 = ((AZ_ENERGY_POINT2H << 8) AZ_ENERGY_POINT2L)
15:0	AZ_ENERGY_POINT3[15:0]	Z-axis focus point 3: Acceleration spectrum energy AZ_ENERGY_POINT3 = ((AZ_ENERGY_POINT3H << 8) AZ_ENERGY_POINT3L)
15:0	AZ_ENERGY_POINT4[15:0]	Z-axis focus point 4: Acceleration spectrum energy AZ_ENERGY_POINT4 = ((AZ_ENERGY_POINT4H << 8) AZ_ENERGY_POINT4L)
15:0	AZ_ENERGY_POINT5[15:0]	Z-axis focus point 5: Acceleration spectrum energy AZ_ENERGY_POINT5 = ((AZ_ENERGY_POINT5H << 8) AZ_ENERGY_POINT5L)
15:0	AZ_ENERGY_POINT6[15:0]	Z-axis focus point 6: Acceleration spectrum energy AZ_ENERGY_POINT6 = ((AZ_ENERGY_POINT6H << 8) AZ_ENERGY_POINT6L)
15:0	AZ_ENERGY_POINT7[15:0]	Z-axis focus point 7: Acceleration spectrum energy AZ_ENERGY_POINT7 = ((AZ_ENERGY_POINT7H << 8) AZ_ENERGY_POINT7L)

15:0	AZ_ENERGY_POINT8[15:0]	Z-axis focus point 8: Acceleration spectrum energy AZ_ENERGY_POINT8 = ((AZ_ENERGY_POINT8H << 8) AZ_ENERGY_POINT8L)
15:0	AZ_ENERGY_AREA1[15:0]	Z-axis focus band 1 acceleration spectrum energy AZ_ENERGY_AREA1=((AZ_ENERGY_AREA1H<< 8 AZ_ENERGY_AREA1L)
15:0	AZ_ENERGY_AREA2[15:0]	Z-axis focus band 2 acceleration spectrum energy AZ_ENERGY_AREA2=((AZ_ENERGY_AREA2H<< 8 AZ_ENERGY_AREA2L)
15:0	AZ_ENERGY_AREA3[15:0]	Z-axis focus band 3 acceleration spectrum energy AZ_ENERGY_AREA3=((AZ_ENERGY_AREA3H<< 8 AZ_ENERGY_AREA3L)
15:0	AZ_ENERGY_AREA4[15:0]	Z-axis focus band 4 acceleration spectrum energy AZ_ENERGY_AREA4=((AZ_ENERGY_AREA4H<< 8 AZ_ENERGY_AREA4L)
15:0	AZ_ENERGY_AREA5[15:0]	Z-axis focus band 5 acceleration spectrum energy AZ_ENERGY_AREA5=((AZ_ENERGY_AREA5H<< 8 AZ_ENERGY_AREA5L)
15:0	AZ_ENERGY_AREA6[15:0]	Z-axis focus band 6 acceleration spectrum energy AZ_ENERGY_AREA6=((AZ_ENERGY_AREA6H<< 8 AZ_ENERGY_AREA6L)
15:0	AZ_ENERGY[15:0]	Total energy of the Z-axis acceleration spectrum AZ_ENERGY = ((AZ_ENERGYH << 8 AZ_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (Z-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AZ_ENERGY_POINT1H AZ_ENERGY_POINT1L CRCH CRCL

AZ_ENERGY_POINT1[15:0]=(((short)AZ_ENERGY_POINT1H <<8)|AZ_ENERGY_POINT1L);

6.2. CAN protocol

6.2.1. Active output format

- The data is sent in hexadecimal format, not ASCII.
- Each data element is transmitted sequentially as a low-byte and a high-byte, which are combined to form a signed short. For example, in the case of data DATA1, DATA1L is the low-byte and DATA1H is the high-byte. The conversion method is as follows: Assuming DATA1 is

the actual data, DATA1H is its high-byte portion, and DATA1L is its low-byte portion, then: $DATA1 = (\text{short})(\text{short}) DATA1H \ll 8 \mid DATA1L$). It is important to note that DATA1H must first be converted to a signed short before shifting, and DATA1 must also be a signed short to represent negative numbers.

Protocol header	Data content	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	High 8 bits of data
0x55	TYPE 【1】	DATA1L[7:0]]	DATA1H[15:8]]	DATA2L[7:0]]	DATA2H[15:8]]	DATA3L[7:0]]	DATA3H[15:8]]

【1】 TYPE (data content):

TYPE	Remark
0x50	time
0x51	acceleration
0x55	temperature
0x5B	Vibration speed
0x5C	Vibration displacement
0x5D	Vibration frequency

6.2.2. Time output

0x55	0x50	YY	MM	DD	HH	MN	SS
name	describe	Remark					
YY	Year						
MM	moon						
DD	day						
HH	hour						
MN	point						

SS	Second	
----	--------	--

6.2.3. Acceleration output

0x55	0x51	AxL	Ax	AyL	AHr	AHr	AHr
name	describe	Remark					
AxL	Acceleration AxL lower 8 bits	Acceleration X = ((AxH<<8) AxL)/32768*16g (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)					
Ax	Acceleration AxH high 8 bits						
AyL	Acceleration AyL lower 8 bits	Acceleration Y = ((AyH<<8) AyL)/32768*16g (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)					
AHr	Acceleration AyH high 8 bits						
AHr	Acceleration AzL low 8 bits	Acceleration Z = ((AzH<<8) AzL)/32768*16g (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)					
AHr	Acceleration AzH high 8 bits						

6.2.4. Temperature output

0x55	0x55	TempL	TempH				
name	describe	Remark					
TempL	Acceleration TempL lower 8 bits	Temperature X = ((TempH << 8) TempL) / 100 °C					
TempH	Acceleration TempH high 8 bits						

6.2.5. Vibration velocity output

0x55	0x5B	VxD	VxD	VxD	Vx	V L	VxD
------	------	-----	-----	-----	----	-----	-----

name	describe	Remark
VxD	Vibration velocity VxL lower 8 bits	Vibration velocity VX (mm/s) = ((VxH <<8) VxL) /100
VxD	Vibration speed VxH high 8 bits	
VxD	Vibration velocity VyL lower 8 bits	Vibration velocity VY (mm/S) = ((VyH <<8) VyL) /100
Vx	Vibration speed VyH high 8 bits	
V L	Vibration speed VzL lower 8 bits	Vibration velocity VZ (mm/s) = ((VzH <<8) VzL) /100
VxD	Vibration speed VzH high 8 bits	

6.2.6. Vibration displacement output

0x55	0x5C	DxL	DxD	DyL	HkDJ	DzL	HkDJ
name	describe	Remark					
DxL	Vibration displacement DxL lower 8 bits	Vibration displacement DX (um) = ((DxH <<8) DxL)					
DxD	Vibration displacement DxH high 8 bits						
DyL	Vibration displacement DyL lower 8 bits	Vibration displacement DY (um) = ((DyH <<8) DyL					
HkDJ	Vibration displacement DyH high 8 bits						
DzL	Vibration displacement						

	DzL lower 8 bits	Vibration displacement DZ (um) =((DzH <<8) DzL)
HkDJ	Vibration displacement DzH high 8 bits	

6.2.7. Vibration frequency output

0x55	0x5D	HkDJ	HkDJ	HkDJ	HkDJ	HkDJ	HZD
name	describe	Remark					
HkDJ	Vibration frequency HZxL low 8 bits	Vibration frequency HZ X (Hz) =((HZxH <<8) HZxL)/10					
HkDJ	Vibration frequency HZxH high 8 bits						
HkDJ	Vibration frequency HZyL lower 8 bits	Vibration frequency HZ Y (Hz) =((HZyH <<8) HZyL)/10					
HkDJ	Vibration frequency HZyH high 8 bits						
HkDJ	Vibration frequency HZzL low 8 bits	Vibration frequency HZ Z (Hz) =((HZzH <<8) HZzL)/10					
HZD	Vibration frequency HZzH high 8 bits						

6.2.8. Write register format

- The data is sent in hexadecimal format, not ASCII.
- All settings require unlocking the register (KEY) first.
- Each register address and write data are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, and the high and low bits of the write data are represented by DATA_H and DATA_L.

Protocol header	Protocol header	register	Data lower 8 bits	High 8 bits of data
-----------------	-----------------	----------	-------------------	---------------------

0xFF	0xAA	ADDR	DATAL[7:0]	DATAH[15:8]
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6.2.9. KEY (unlock)

Register Name: KEY

Register address: 105 (0x69)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	KEY[15:0]	Unlock register: When performing a write operation, you need to set this register first

Example: To unlock, write 0xB588 to this register (other values are invalid)

FF AA 69 88 B5

6.2.10. SAVE (Save/Reboot/Reset to Factory)

Register Name: SAVE

Register address: 0 (0x00)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Reboot: 0x00FF Factory Reset: 0x0001

Example: FF AA 00 FF 00 (Reboot)

6.2.11. BAUD-CAN (CAN baud rate)

Register Name: BAUD_CAN

Register address: 5 (0x05)

Read/write direction: R/W

Default value: 0x0004

Bit	NAME	FUNCTION
15:4		
3:0	BAUD_CAN[3:0]	Set the CAN baud rate: 0000(0x00): 1000000bps 0001(0x01): 800000bps 0010(0x02): 500000bps 0011(0x03): 400000bps 0100(0x04): 250000bps 0101(0x05): 200000bps 0110(0x06): 125000bps 0111(0x07): 100000bps 1000(0x08): 80000bps 1001(0x09): 50000bps 1010(0x0A): 40000bps 1011(0x0B): 20000bps 1100(0x0C): 10000bps 1101(0x0D): 5000bps 1110(0x0E): 3000bps

Example: FF AA 05 02 00 (set the CAN baud rate to 500000)

6.2.12. RRATE (output rate)

Register Name: RRATE

Register address: 3 (0x03)

Read/write direction: R/W

Default value: 0x0006

Bit	NAME	FUNCTION
15:4		
3:0	RRATE[3:0]	Set the output rate: 0001(0x01): 0.2Hz 0010(0x02): 0.5Hz 0011(0x03): 1Hz 0100(0x04): 2Hz 0101(0x05): 5Hz 0110(0x06): 10Hz 0111(0x07): 20Hz 1000(0x08): 50Hz 1001(0x09): 100Hz 1011(0x0B): 200Hz

Example: FF AA 03 03 00 (set 1Hz output)

6.2.13. IICADDR (device address)

Register Name: IICADDR

Register Address: 26 (0x1A)

Read/write direction: R/W

Default value: 0x0050

Bit	NAME	FUNCTION
15:8		

7:0	IICADDR [7:0]	Set the device address for I2C and Modbus communication 0x01~0x7F
Example: FF AA 1A 02 00 (set the device address to 0x02)		